



Anki Vector

The Anki Vector is the perfect robotic "sidekick" for your smarthome - and it has a personality!
<https://dgit.in/sep18-20-1>



Peak Social Media

All around the world, the number of daily active users on social media platforms is reducing
<https://dgit.in/sep18-20-2>

Unusual Computers

While most computers we know of today are semiconductor based, there are a number of wacky computers out there.

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CLOCKWORK

The Antikythera mechanism is often considered the oldest computer, and it was used for astronomical calculations. This mysterious object is dated to about the end of the 2nd century BC. The device predates the next known clockwork mechanism by almost a thousand years. The astrolabe, invented in the same time period, was a more primitive instrument, but was essentially a computing device for solving astronomical problems. Astronomical calculations for navigation was the driving force behind many of the early mechanical computers.

In 1642, Pascal designed the mechanical calculator known as the Pascaline. The machine could be used to add and subtract numbers. Leibniz's stepped reckoner improved on this version, and added the functionality for multiplication and division directly, instead of repeatedly performing an operation. Charles Babbage made a number of advanced computers including the Difference Engine to calculate polynomials and the



The clockwork rover by NASA

Analytical Engine, which was a general purpose computer. The Analytical Engine could be programmed through punch cards then used for looms, had a memory storage unit, and also an arithmetic unit.

Some of the mechanical computers used in wartime were actually superior to electronic alternatives. They were more accurate, and the calculations were done in real time, instead of the clock cycles and were even used to target moving ships. One of these, the Mark 1A, Fire Control Computer computed the angles at which the ships should fire their guns, the velocities of the projectiles, and their time of flight, factoring in wind, gravity and parallax. However, due to space constraints, the military eventually embraced electronic computers.

For most of human history, we have actually been using mechanical computers. Now mechanical comput-

ers can be both analog and digital. Modern computers are electronic digital computers. Think of the difference in this way, a mechanical analog clock has moving hands to display the time, even if the numerals are not imprinted on the face. A mechanical digital clock would show the actual numbers on a cylinder moved by gears.

While electronic digital computers might have become ubiquitous now, mechanical computers are making a comeback. The approach combines nanotechnology and the design principles of the earliest mechanical computers, such as the Difference Engine. This is because on a nanoscale, mechanical motion is very rapid, and are a low cost option to existing microprocessors. These nanomechanical computers can be integrated with appliances and toys - things that need computation, but do not need to do it very quickly. These nanomechanical